3.3 Needs of gear box

1. For reducing the engine speed and increasing the torque.
2. For reducing the engine torque and increasing speed.
3. For the selection of speed levels according to the driver's requirement.
4. For pushing the vehicle in the backward direction.
5. For neutralizing the vehicles when there is no need to transmit the engine power.

3.4 Principle of gear box mechanism

Gear box works on the principle of lever mechanism.

3.5 Types of gear box

The following are the gear boxes used in automobiles,

1. Sliding mesh gear box
2. Constant mesh gear box
3. Synchromesh gear box
4. Epicyclic gear box
5. Automatic gear box

There are many types of gear boxes available. The important gear boxes are discussed below.

3.5.1 Sliding mesh gear box

Sliding mesh gear box was used in earlier day's vehicles. Figure 3.5.1 presents the details of the sliding mesh gear box. In this gear box spur gears are present. This box is suitable for heavy vehicles. By sliding the gears this type of gearboxes move the gears and mesh them and hence it is called as sliding mesh gear box.

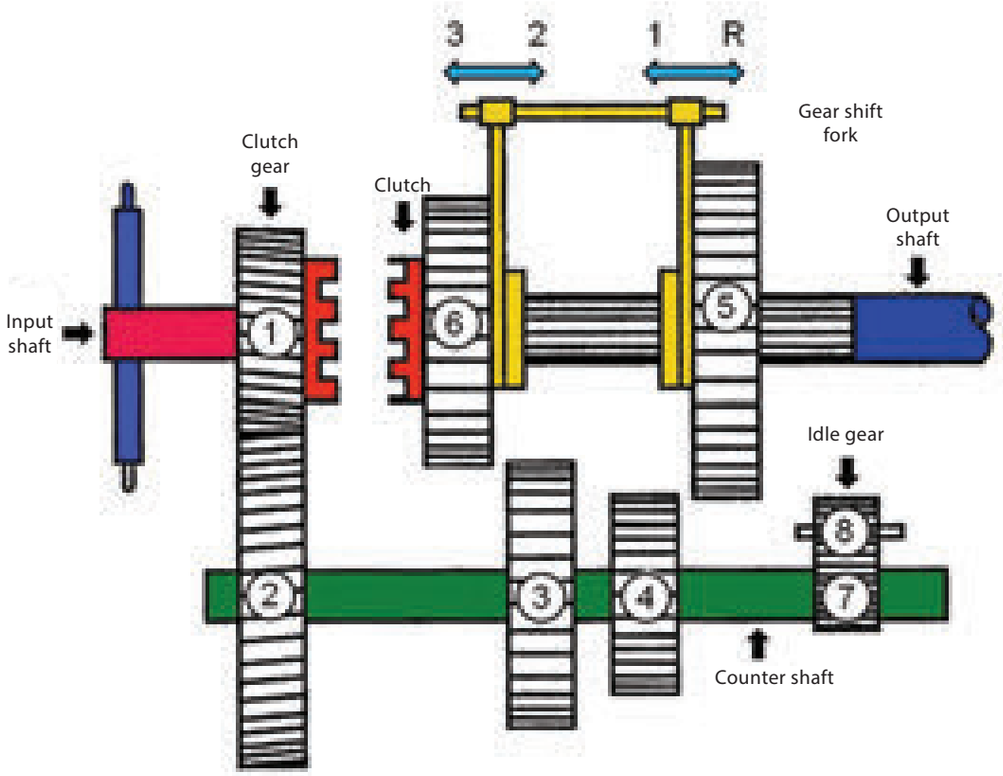


Figure 3.5.1 Sliding mesh gear box

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